

Representation Theory: A First Course - Self Study

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Abstract

This text contains my notes on Representation Theory: A First Course, by William Fulton and Joe Harris. I read this book with a friend, starting mid spring semester of my sophomore year, 2025. My notes largely paraphrase the text, and I work out whatever examples I see fit. Since I had no knowledge of differentiable manifolds or Lie groups coming into this book, I have complimented my reading with other sources to ensure understanding, such as [insert sources when I use them lol]

Contents

I	Finite Groups	2
1	Representations of Finite Groups	3
1.1	Definitions	3

Part I

Finite Groups

Lecture 1

Representations of Finite Groups

1.1 Definitions

Definition 1.1: Representation

A representation of a finite group G on a finite-dimensional complex vector space V is a group homomorphism $\rho : G \rightarrow \mathrm{GL}(V)$ from G to the group $\mathrm{GL}(V) = \mathrm{Aut}_{\mathbb{C}\text{-}\mathbf{Vect}}(V)$. When there is little ambiguity about ρ , we simply refer to V as a representation of G , and we write gv for $\rho(g)(v)$. We often call the dimension of V the *degree* of ρ . We say ρ gives V the structure of a G -module.

If V, W are representations of G , then a morphism of representations is a $\mathbb{C}$ -linear map $\phi : V \rightarrow W$ such that

$$\begin{array}{ccc} V & \xrightarrow{\phi} & W \\ g \downarrow & & \downarrow g \\ V & \xrightarrow{\phi} & W \end{array}$$

commutes for all $g \in G$. That is, $\phi(gv) = g\phi(v)$ for all $g \in G$ and $v \in V$. We call such a map G -linear. It follows immediately that $\ker \phi, \mathrm{im} \phi, \mathrm{coker} \phi$ are all G -modules.

Definition 1.2: Subrepresentation

A subrepresentation W of a representation V is a subspace $W \subseteq V$ that is G -invariant; that is, g is an automorphism of W as well as V .

Some common representations are the direct sum $V \oplus W$, given by $g(v, w) = (gv, gw)$, and the tensor product $V \otimes W$, given by $g(v \otimes w) = gv \otimes gw$. This generalizes to tensor powers $V^{\otimes n}$, and we can view the exterior power $\bigwedge^n(V)$ and the symmetric power $\mathrm{Sym}^n(V)$ as subrepresentations of it.

The dual $V^* = \mathrm{Hom}(V, \mathbb{C})$ is also a representation. Let $\rho : G \rightarrow V$ be a representation, and $\rho^* : G \rightarrow V^*$ be the dual. We want the representations to respect the natural map

$$\langle \rho^*(g)(v^*), \rho(g)(v) \rangle$$

for all $g \in G$, $v \in V$, and $v^* \in V^*$. This definition forces us to define gv^* as the map $(gv^*)(x) = v^*(g^{-1}x)$. The book denotes this as ${}^t\rho(g^{-1})$.

Exercise 1.1

Show that this definition of ρ^* satisfies the above relation.

Proof. The fact that the above relation is satisfied is immediate:

$$\langle gv^*, gv \rangle = (gv^*)(gv) = v^*(g^{-1}gv) = v^*(v).$$

I feel we must also show r^* is indeed a representation. First, we show g is an automorphism of V^* . Surjectivity follows immediately. Let $v^* \in V^*$, and note that since g is an automorphism on V , for every $v \in V$ there exists $w \in V$ such that $gw = v$. Therefore, $v^*(gw) = v^*(v)$, and thus the composition

$$\begin{array}{ccccc} & & v^* & & \\ & \nearrow & & \searrow & \\ V & \xrightarrow{g} & V & \xrightarrow{v^*} & \mathbb{C} \end{array}$$

commutes. Since every g is the inverse of some $g^{-1} \in G$, the given definition is surjective, since $g^{-1}v^* = v^* \circ g \in \text{Hom}(V, \mathbb{C})$.

To show injectivity, we need only show $\ker g = 0$ for all $g \in G$. Indeed, if $gv^* = 0$, then $v^*(g^{-1}x) = 0$ for all $x \in V$. Since g^{-1} is an automorphism of V , it follows that $v^*(x) = 0$ for all $x \in V$ by surjectivity, and thus $v^* = 0$. Therefore, $\rho^* : G \rightarrow \text{GL } V^*$.

Finally, we show ρ^* is a group homomorphism. This is immediate; let $g, h \in G$. We have

$$(gh)v^*(x) = v^*((h^{-1}g^{-1})x) = (hv^*)(g^{-1}x) = g(h(v^*))(x).$$

Therefore, $g \circ h = gh$, and thus $\rho^*(gh) = \rho^*(g) \circ \rho^*(h)$. Therefore, V^* is a representation of G under this definition. ■

need to show $V^* \otimes W \cong \text{Hom}_{\mathbb{C}}(V, W)$.